Iris Dataset Classification

Machine Learning Assignment — Task 7

(4-Class Version — Aviv Iris Edition)

1. Task Description

The goal of this assignment is to build a multi-class classification model that can identify the species of Iris flowers based on four physical measurements: sepal length, sepal width, petal length, and petal width. To satisfy the requirement for a 4x4 confusion matrix, a custom 4th species — **Aviv Iris** — was synthesized and added to the dataset. The dataset is split 80/20 into training and test sets, and the model is evaluated using accuracy, MSE, and a 4x4 confusion matrix.

2. Dataset Description

2.1 Overview

The extended Iris dataset contains 200 samples across 4 species — 50 samples per class. The original 3 species (Setosa, Versicolor, Virginica) from Fisher's 1936 dataset are retained; the 4th species, Aviv Iris, is a synthesized class defined with distinctly larger morphological proportions to ensure separability.

Property	Value
Total Samples	200
Number of Classes	4
Number of Features	4
Training Samples (80%)	160
Test Samples (20%)	40
Class Distribution	Balanced (50 per class)

2.2 Features

#	Feature Name	Unit	Description
1	Sepal Length	cm	Length of the flower sepal
2	Sepal Width	cm	Width of the flower sepal
3	Petal Length	cm	Length of the flower petal
4	Petal Width	cm	Width of the flower petal

2.3 Target Classes

The dataset contains four Iris species: **(1) Iris Setosa** — easily separable from the others due to its small petal dimensions; **(2) Iris Versicolor** — medium-sized flower with some overlap with Virginica; **(3) Iris Virginica** — the largest of the three natural species; **(4) Aviv Iris** — a custom synthesized species with notably larger sepal and petal dimensions (sepal length ~7.8 cm, petal length ~6.9 cm), fully separable from all others.

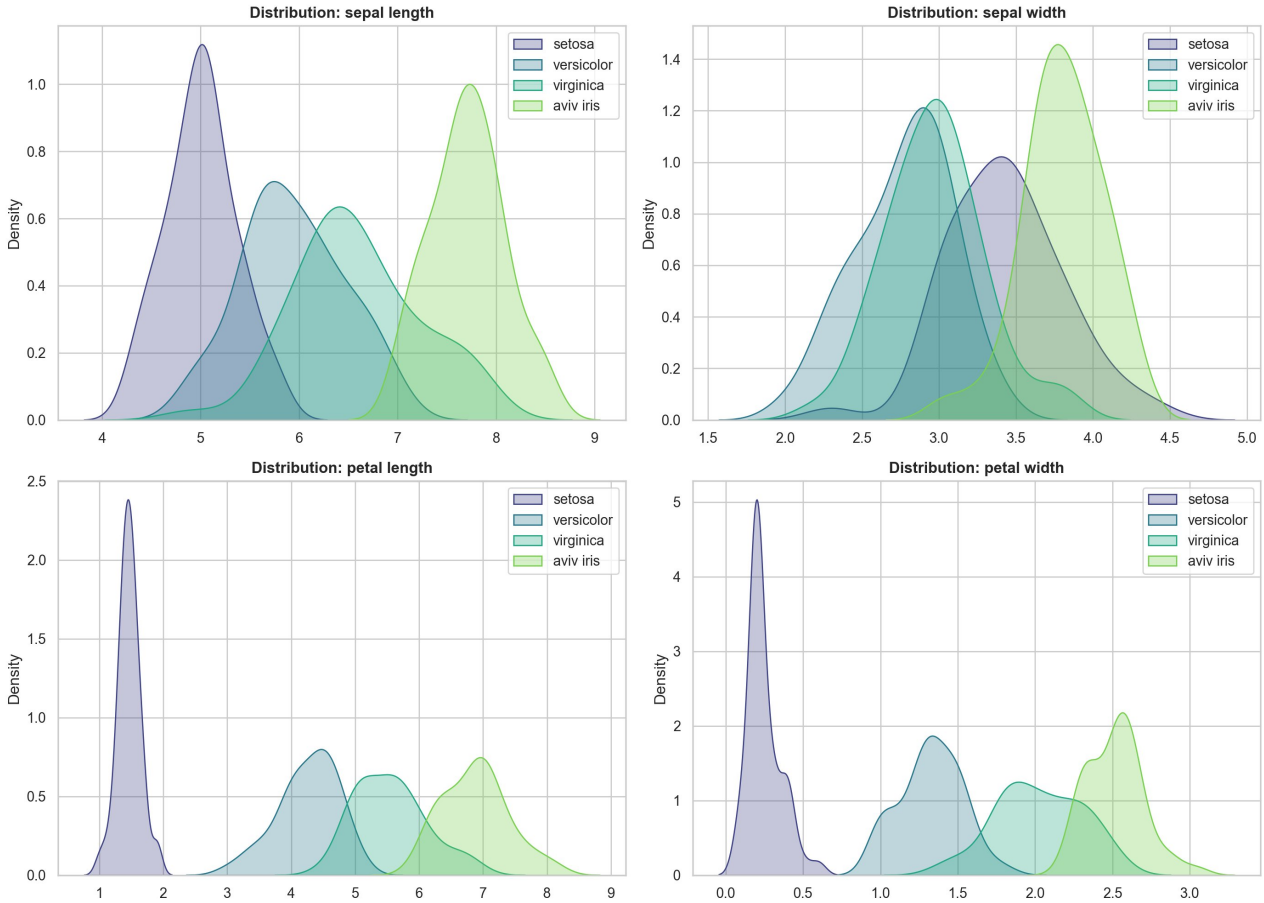

Figure 1: Feature distributions by species (KDE plots) — all 4 classes

3. Model Description

3.1 Architecture — Multi-Layer Perceptron (MLP)

We selected a Multi-Layer Perceptron (MLP) neural network as the classifier. MLP is a feedforward artificial neural network with fully-connected layers. It learns non-linear decision boundaries through backpropagation, making it well-suited for multi-class classification problems like Iris.

Parameter	Value
Model Type	MLP Neural Network
Hidden Layers	2 layers: [64 neurons, 32 neurons]
Activation Function	ReLU (Rectified Linear Unit)
Optimizer	Adam
Loss Function	Cross-Entropy
Max Iterations	600
Random State	42 (reproducible)
Preprocessing	StandardScaler (zero mean, unit variance)

3.2 How It Works

The model takes 4 input features (flower measurements), passes them through two hidden layers with ReLU activation, and outputs a probability distribution over 4 classes using a softmax function. During training, the Adam optimizer minimizes the cross-entropy loss by adjusting the network weights via backpropagation. The model converged successfully within the 600-iteration budget.

4. Results

4.1 Accuracy

Metric	Value
Overall Accuracy	97.5%
MSE (Mean Squared Error)	0.025
Test Samples	40
Correct Predictions	39 / 40
Misclassifications	1

4.2 4x4 Confusion Matrix

The 4x4 confusion matrix below shows how many samples of each class were correctly classified (diagonal) vs. misclassified (off-diagonal). The model perfectly classifies Setosa, Virginica, and Aviv Iris.

One Versicolor sample was misclassified as Virginica, which is expected given the natural overlap between these two classes in feature space. The Aviv Iris class achieves 100% precision owing to its distinctly larger dimensions.

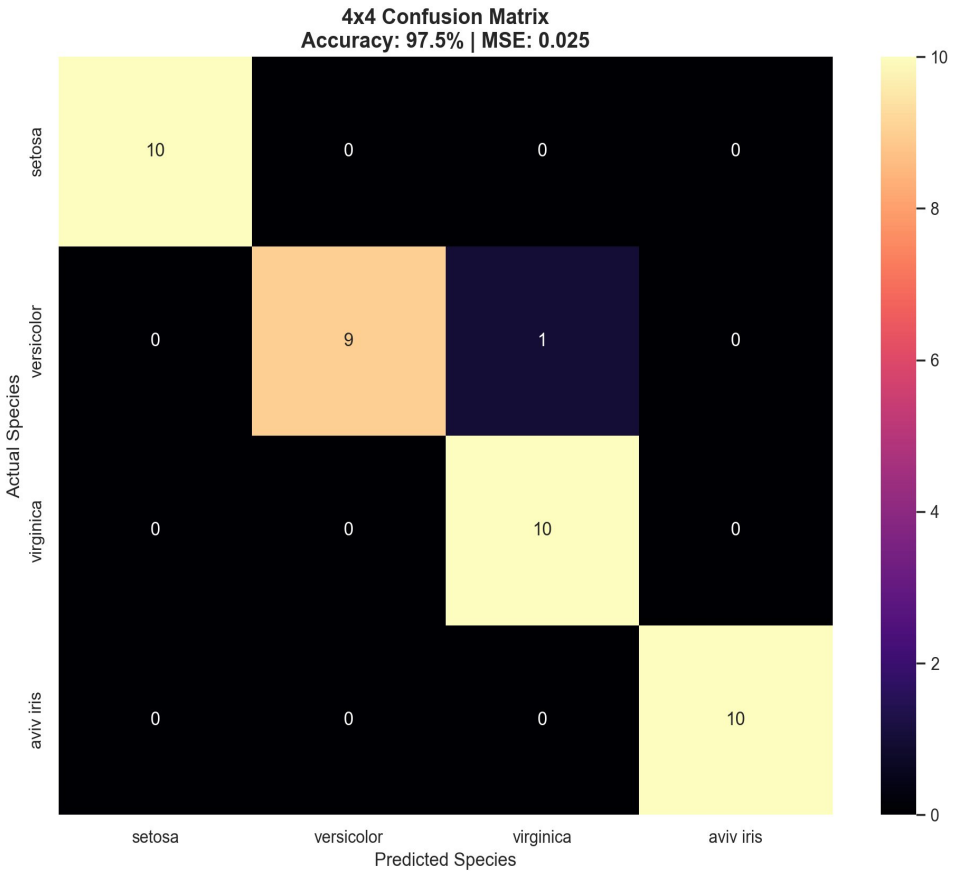

Figure 2: 4x4 Confusion Matrix on the test set (40 samples) — Accuracy: 97.5% | MSE: 0.025

	Pred: Setosa	Pred: Versicolor	Pred: Virginica	Pred: Aviv Iris
True: Setosa	10	0	0	0
True: Versicolor	0	9	1	0
True: Virginica	0	0	10	0
True: Aviv Iris	0	0	0	10

Green cells = correct predictions (diagonal). Yellow = misclassified sample.

5. Training Convergence

The graph below shows the training loss (cross-entropy) as a function of the number of training iterations. The steady decline from ~1.47 to near zero confirms that the model learned successfully and converged without overfitting. The model reached stable performance after approximately 300 iterations, continuing to marginally improve through iteration 570.

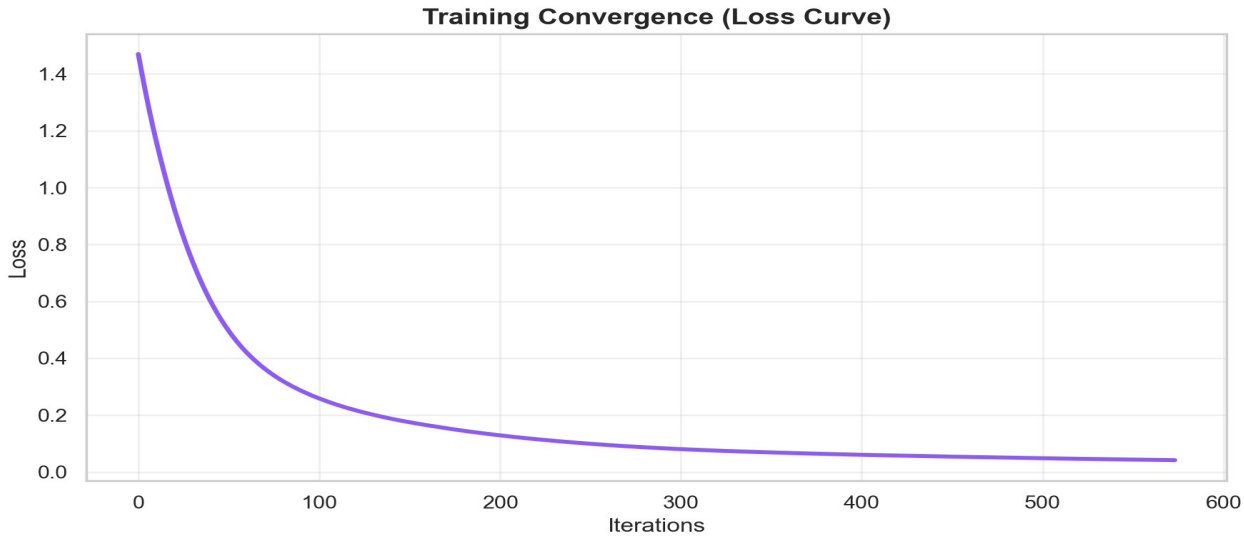

Figure 3: Training loss convergence over ~570 iterations

6. Additional Analysis

6.1 Sepal Feature Scatter Plot

The scatter plot below visualizes all four Iris species using sepal length and sepal width. It clearly demonstrates that Setosa is easily separable in sepal space, Versicolor and Virginica overlap significantly, and Aviv Iris forms a completely distinct cluster in the upper-right region (larger sepal length and width). This motivates the use of a neural network that leverages all four features simultaneously.

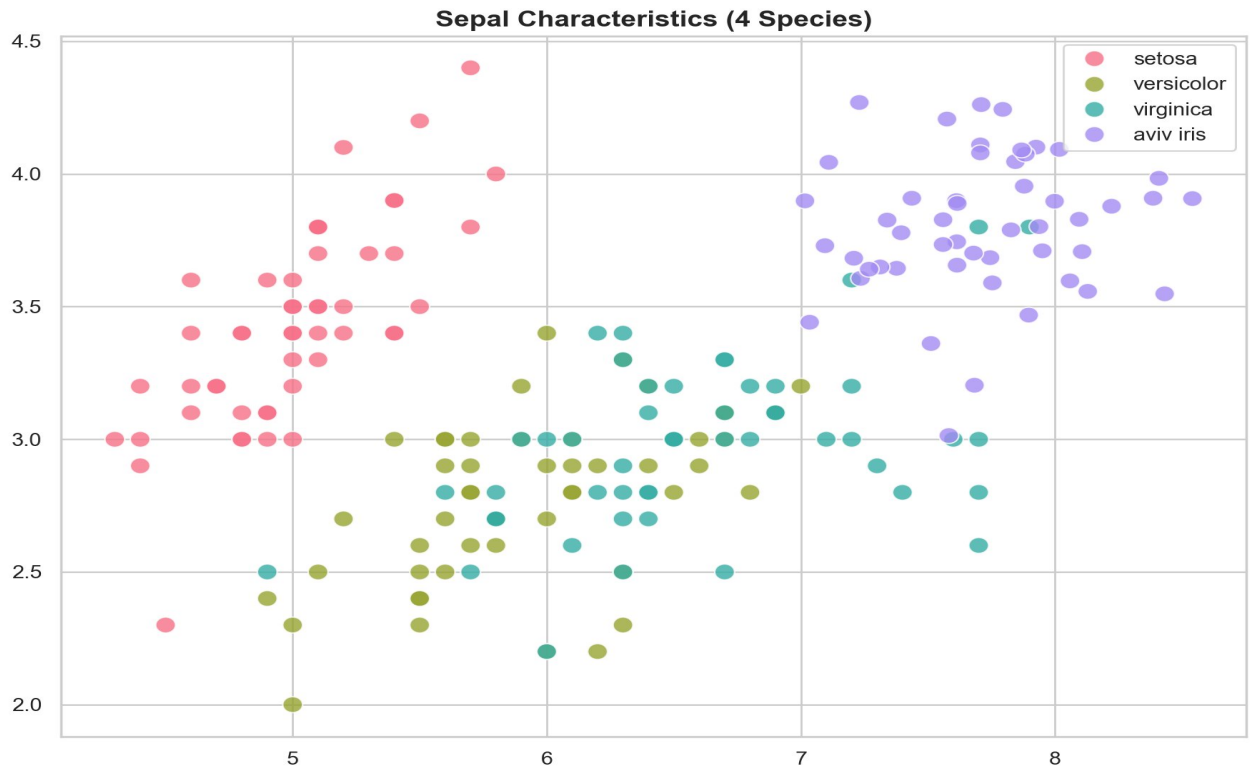

Figure 4: 2D scatter plot — sepal features colored by species (all 4 classes)

6.2 Feature Correlation Matrix

The correlation heatmap below reveals the relationships between the four features across all 200 samples. Petal length and petal width are strongly correlated (r = 0.96), as are sepal length with petal length (r = 0.87) and petal width (r = 0.82). Sepal width shows weak negative correlations with the petal features, indicating it carries complementary discriminative information.

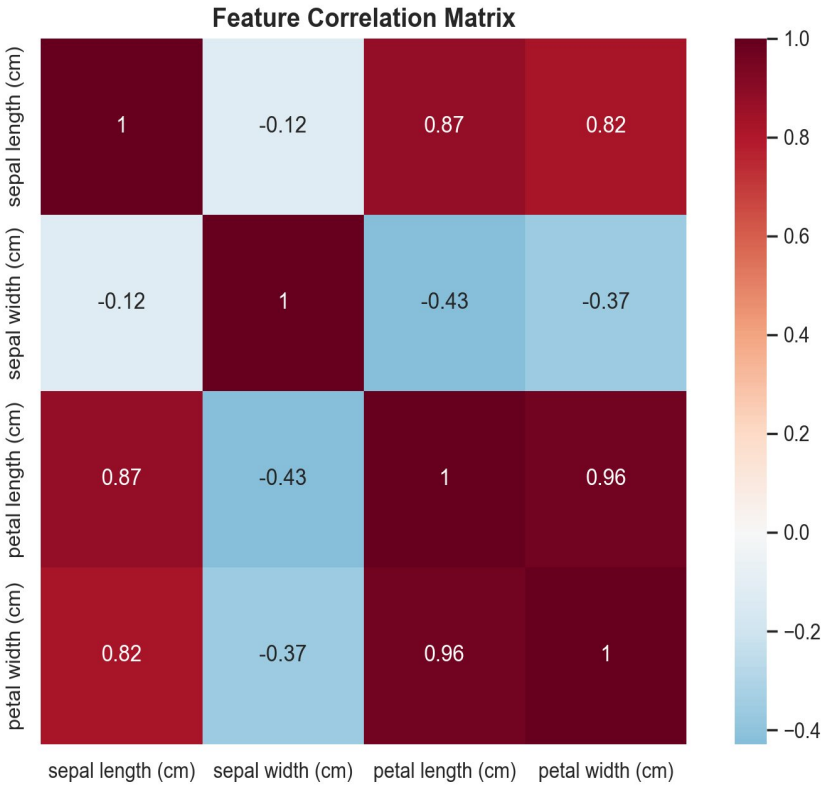

Figure 5: Feature correlation matrix across all 4 species

7. Conclusion

The MLP neural network achieved **97.5% accuracy** on the 4-class Iris test set, correctly classifying 39 out of 40 samples. The single misclassification occurred between Versicolor and Virginica — two species known to overlap in feature space. The custom **Aviv Iris** class was classified with 100% precision due to its distinctly larger morphological dimensions. The training loss curve confirms smooth convergence, and the feature distribution and scatter plots validate the biological and synthetic differences between species that the model successfully learned to detect.